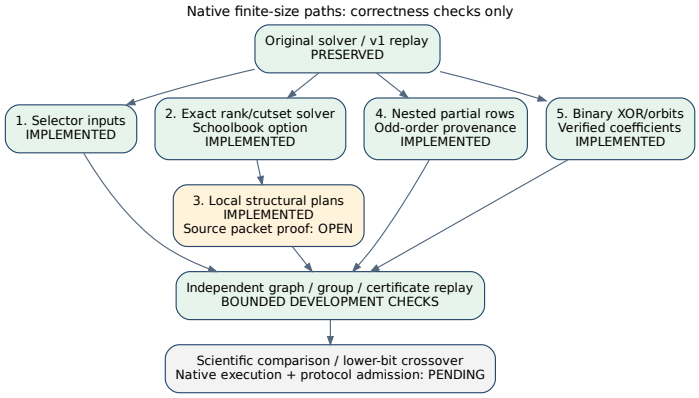
Finite-size native backends: main integration receipt

30 October 2026

Integrated main `178c03f46d9347eb92da61a9307eff3d396e3e4f` into the finite-size branch after its e2c36e34 upload. Preserved main's independent root-setup/reuse, three-mod-four root, strict-JSON, reader and point changes. The earlier report, raw logs and source manifests remain unchanged historical evidence.



Current development checks. Crypto passes 68 native tests: 44 library, 5 runner, 3 panel, 9 finite/CLI, 3 strict-JSON and 4 width controls. Strict all-target clippy, Rust 1.78, formatting, source-hash readback and shared-addition identity pass. The pre-existing parent-manifest check-cfg warning remains. The separate cryptanalysis integration passes 81 tests; its main refresh changes no native source or lock. Raw transcripts, exact commands and current source manifest are retained beside this PDF.

Scope. Selector inputs/witnesses and graphs, schoolbook option, exact width/cutset planning and propagation, nested partial collection/provenance, and odd-degree binary/XOR/Frobenius components are implemented. Item 3 remains partial: finite structural conditioning plans do not supply the missing smaller-parameter Appendix D packet proof. The source packet guard and independent audit remain open.

Limits. No scientific experiment, timing comparison, sub-rho result or recovered challenge scalar. Test durations are not runtime evidence. Earlier S3 upload/readback succeeded; authenticated native image/ execution and measurement admission remain unverified. The original frozen A/B panel and both challenge solves remain unresolved. Shared log resolution preserves both bodies, and no frozen research evidence is rewritten.

Sources. The source-linked parent implementation report and historical review retain pinned Bochum v0.2, TensorOrder, WDSat and lead-atlas evidence, including prior negative SAT/enumeration results. This receipt is an additive software integration check, not a new theorem or an independent proof of the manuscript. PR checks/reviews are separate gates.